HEMANTH P P

Software Engineer | AI / ML · Full-Stack Development · Cloud Infrastructure · Multi-Agent Orchestration

Tel +91 8078006591 @ hemanthpp1234@gmail.com in linkedin.com/in/hemanth-p-p-277930304 gh github.com/Hemanthpp web Portfolio

Tharipakkunil House, Tholambra P.O, Kannur, Kerala · Open to On-site / Remote / Hybrid

SUMMARY

B.Tech Information Technology student (CUSAT, 2023–Present) with proven hands-on experience across the full software engineering stack — AI/ML model development, backend systems, full-stack web applications, cloud deployment, and multi-agent orchestration. Built **RepoPilot-AI**, a production-grade autonomous multi-agent system (Google ADK 2.0 + Gemini API) with 6 LLM sub-agents, custom tool pipelines, FastAPI serving, and a comprehensive Pytest suite. Interned at three companies across AI/Computer Vision, frontend engineering, and cloud ML. Certified in AI/ML (NPTEL, 81%), Cloud Computing (IBM), and DevOps on AWS. Proficient in Python, JavaScript/TypeScript, SQL, REST APIs, Docker, Git/CI-CD, and multiple ML frameworks. Strong communicator in remote and on-site settings; fast learner with a systematic problem-solving mindset.

WORK EXPERIENCE

Software Developer Intern — AI & Computer Vision
Jun 2025

Techgentsia Software Technologies Pvt. Ltd

Cherthala Infopark, Kerala

- Designed and deployed a YOLO-based object detection pipeline — handled data preprocessing, model configuration, hyperparameter tuning, and inference optimization within a 3-week sprint.
- Built an end-to-end integration spanning Python backend inference and frontend visualization; debugged cross-layer failures by analyzing logs, tracing request flows, and documenting root causes with runbook-style notes.
- Adapted rapidly to an unfamiliar codebase across backend, frontend, and ML layers — demonstrated learning velocity, methodical debugging, and delivery under deadline.

AI & ML Intern
Jun 2024 – Jul 2024

CSRBOX & IBM SkillsBuild Internship Program

Remote

- Developed and trained ML models using supervised and unsupervised learning — applied data cleaning, feature selection, and model evaluation using Python, Pandas, NumPy, and Scikit-learn.
- Collaborated across a distributed remote team to build, validate, and present a functional AI/ML prototype against strict quality and documentation standards.

Frontend Development Intern
May 2024 – Jun 2024

Keltron Technologies

Kannur, Kerala

- Built responsive web interfaces (HTML5, CSS3, JavaScript) within a production engineering team using Git-based version control, code review workflows, and cross-browser testing.
- Authored structured implementation notes and modular technical documentation, maintaining code quality standards within an active engineering workplace.

PROJECTS

RepoPilot-AI — Autonomous Multi-Agent Developer Assistant
Python · Google ADK 2.2.0 · Gemini API · FastAPI · Multi-Agent Orchestration · MCP Server · Pytest · Structured Logging

- Architected a production-grade autonomous system orchestrating 6 LLM sub-agents via Google ADK 2.2.0 — designed inter-agent communication, resource management, graceful error handling, and audit logging.
- Built 4 custom tool pipelines (clone_repo, get_tree, run_analysis, run_tests) with precise JSON schemas, subprocess management, and structured input/output contracts.
- Deployed FastAPI model-serving infrastructure with PII scrubbing, injection detection, rate limiting, and a full Pytest test suite — ensuring production readiness and security.

Event Helper Service System — Mobile Application
Flutter · Firebase Authentication · Cloud Firestore · Service Booking Workflows

- Developed a multi-service mobile app for discovering and booking vendors — integrated Firebase Auth and Cloud Firestore for scalable real-time record management and user security.

Student Management System — Responsive CRUD Web App
HTML5 · CSS3 · JavaScript (ES6+) · DOM APIs · LocalStorage

- Engineered a fully responsive student record application without external frameworks — real-time CRUD operations, search filtering, and DOM manipulation via LocalStorage.

Period Tracker & Kit Delivery System — DBMS & OOP Software Architecture
SQL · Entity-Relationship Modelling · OOP System Design · Notification Workflows

- Designed a structured database application comprising four relational entities (Users, PeriodDetails, Kits, KitDelivery) to monitor health records, manage physical inventory, and coordinate scheduled deliveries.
- Implemented OOP modularity (encapsulation, inheritance, polymorphism) and scripted automated email notification triggers, establishing theoretical fluency in process automation workflows.

TECHNICAL SKILLS

LANGUAGES

Python · JavaScript (ES6+) · TypeScript · SQL · C++ · Bash

AI / ML FRAMEWORKS

TensorFlow · PyTorch · Scikit-learn · YOLO (Object Detection) · Google Gemini API · Google ADK 2.0 · LLM APIs · Prompt Engineering

BACKEND & APIS

FastAPI · Django REST · REST APIs · JSON / YAML · MCP Servers

FRONTEND & MOBILE

React · Next.js · Flutter · HTML5 / CSS3 · Responsive Design

DATA & DATABASES

NumPy · PostgreSQL · Firebase Firestore · Data Preprocessing

CLOUD & INFRASTRUCTURE

AWS (Certified) · GCP · Docker · Kubernetes (Basic) · Linux CLI · CI/CD Pipelines

TOOLS & DEVOPS

Git / GitHub · Pytest · Postman · VS Code · Structured Logging

CS FUNDAMENTALS

Data Structures & Algorithms · OOP Design · Networking (HTTP) · OS Concepts

CERTIFICATIONS

AI: Concepts & Techniques — 81%
NPTEL / IISc Bangalore · Jul – Oct 2025

Startup School: Prompt to Prototype
Google for Startups · Dec 2025

5-Day Gen AI Intensive — AI Agents Course
Kaggle × Google · 2025

Fundamentals of DevOps on AWS
Simplilearn / AWS SkillUp · May 2026

Cloud Computing Fundamentals
IBM SkillsBuild · Apr 2026

AI & ML — IBM SkillsBuild Internship
CSRBOX & IBM SkillsBuild · Jul 2024

EDUCATION

B.Tech – Information Technology
Cochin University College of Engineering Kuttanad (CUSAT)
2023 – Present | Kuttanad, Kerala

HSE (Class XII) – Science
GHSS Malur
2020 – 2022 | Malur, Kerala

LANGUAGES

English — Professional

Malayalam — Native